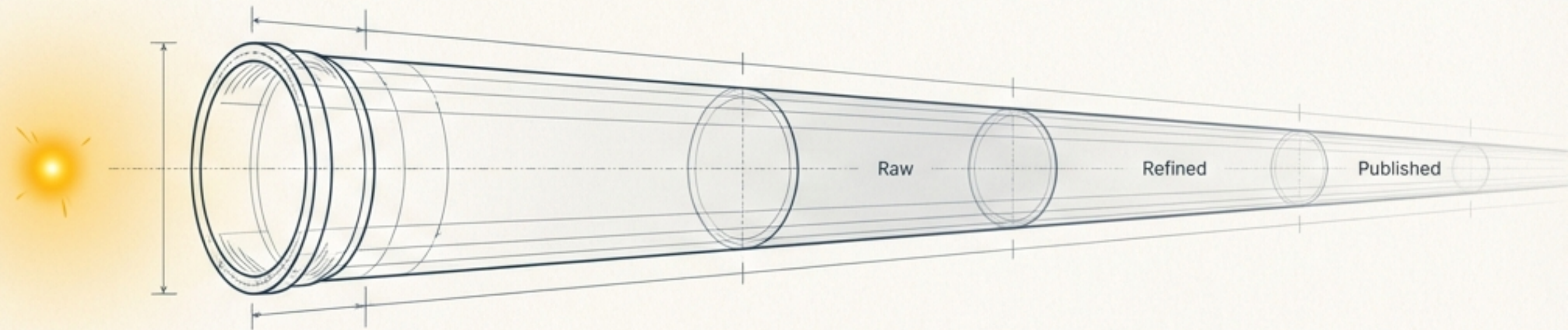


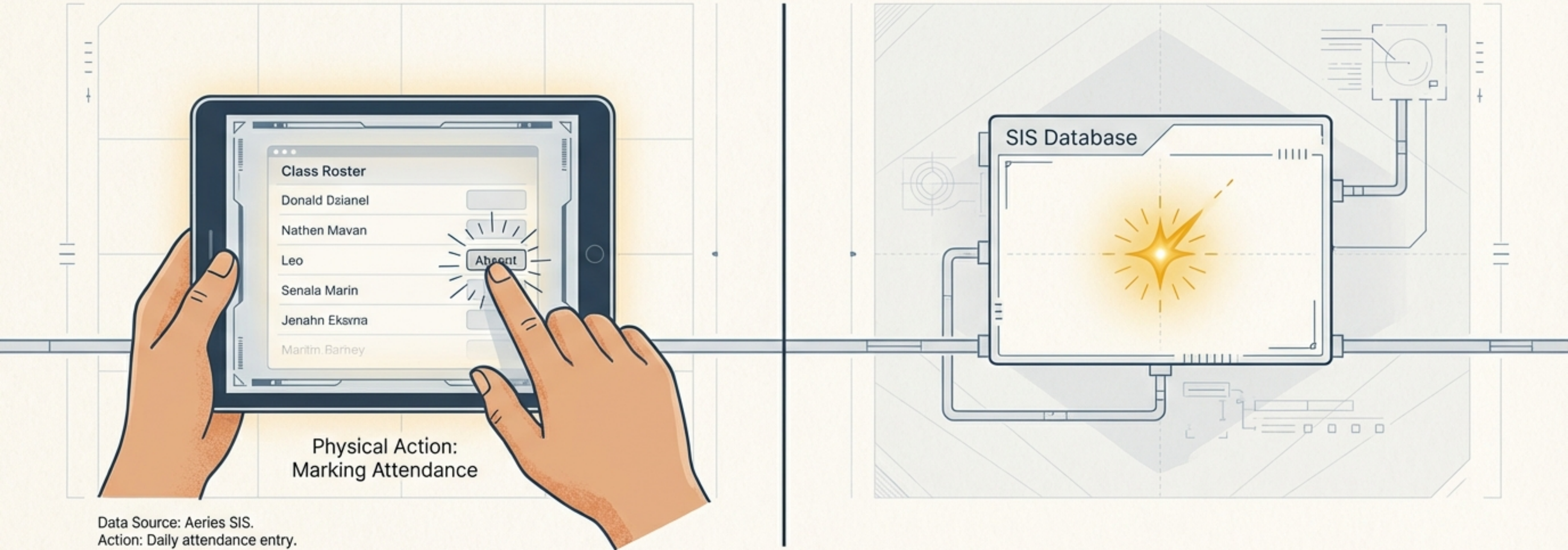
The Secret Life of a Data Point

Following the journey of a single attendance record from a classroom click to a student intervention



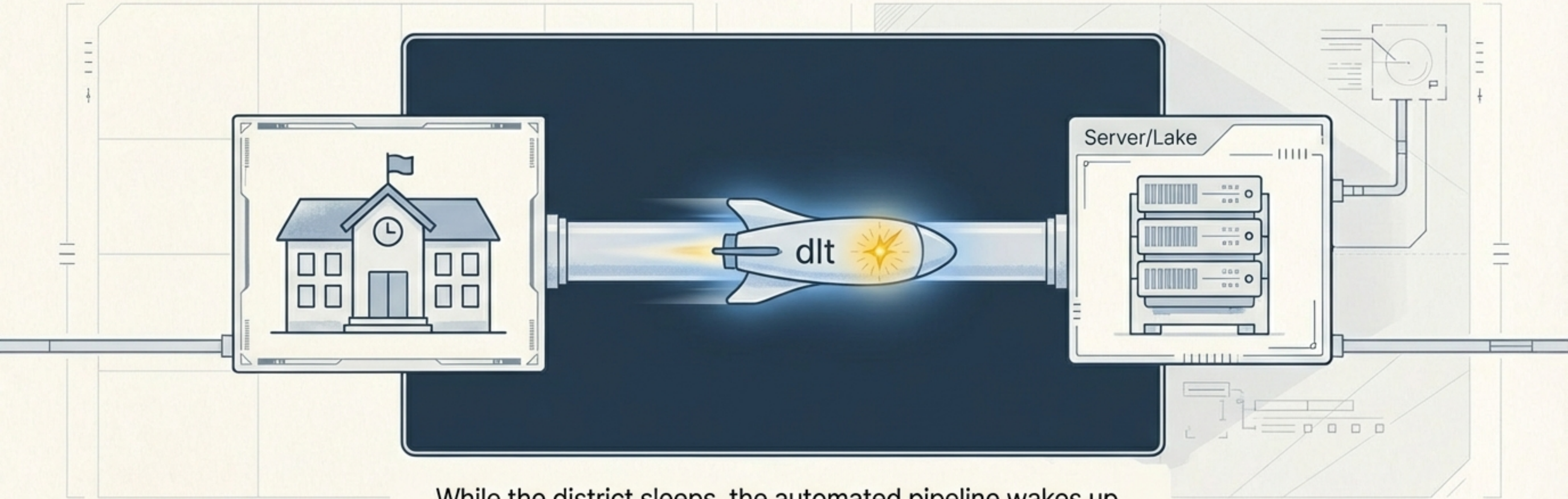
Every day, thousands of decisions are made in our schools. But how does a simple action—marking a student absent—travel through the OSS Framework to become a life-changing insight? This is the story of that journey.

Tuesday, 8:05 AM: It Starts with a Click



The record is currently trapped in the Student Information System (SIS). It is an “island of data”—useful to the teacher for roll call, but invisible to the district for analytics.

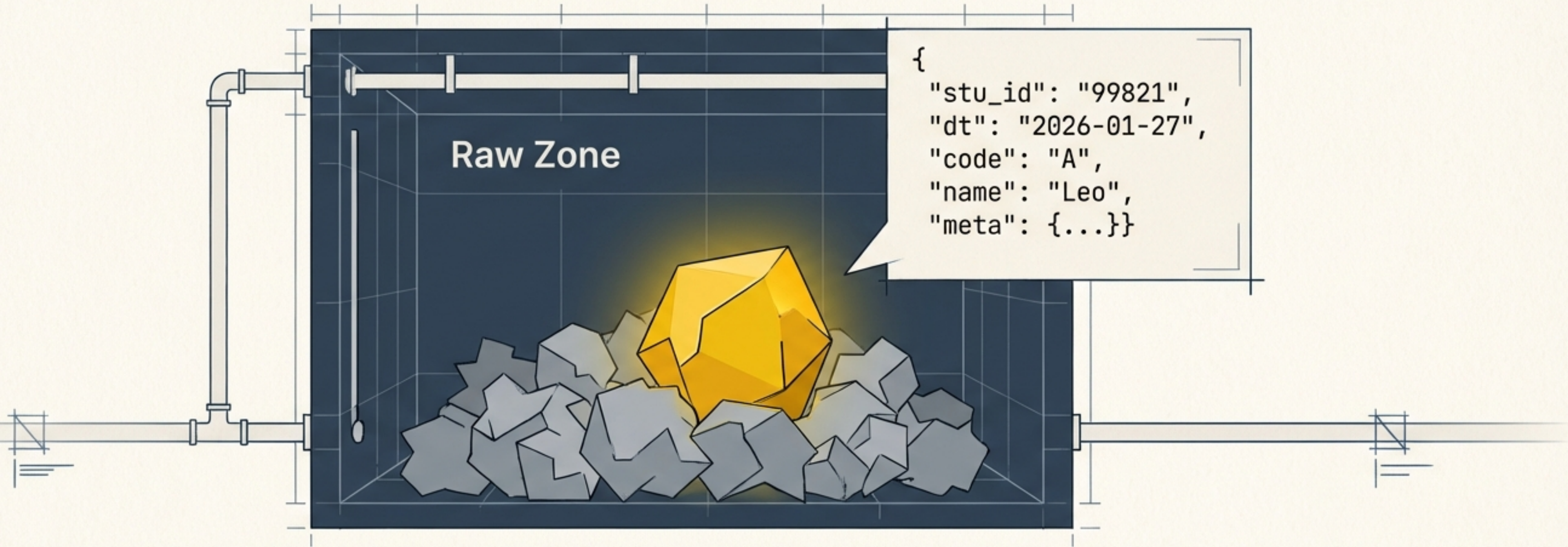
The Night Shift: Crossing the Threshold



While the district sleeps, the automated pipeline wakes up.
The **dlt** (Data Load Tool) **extracts the record via the Aeries API**.

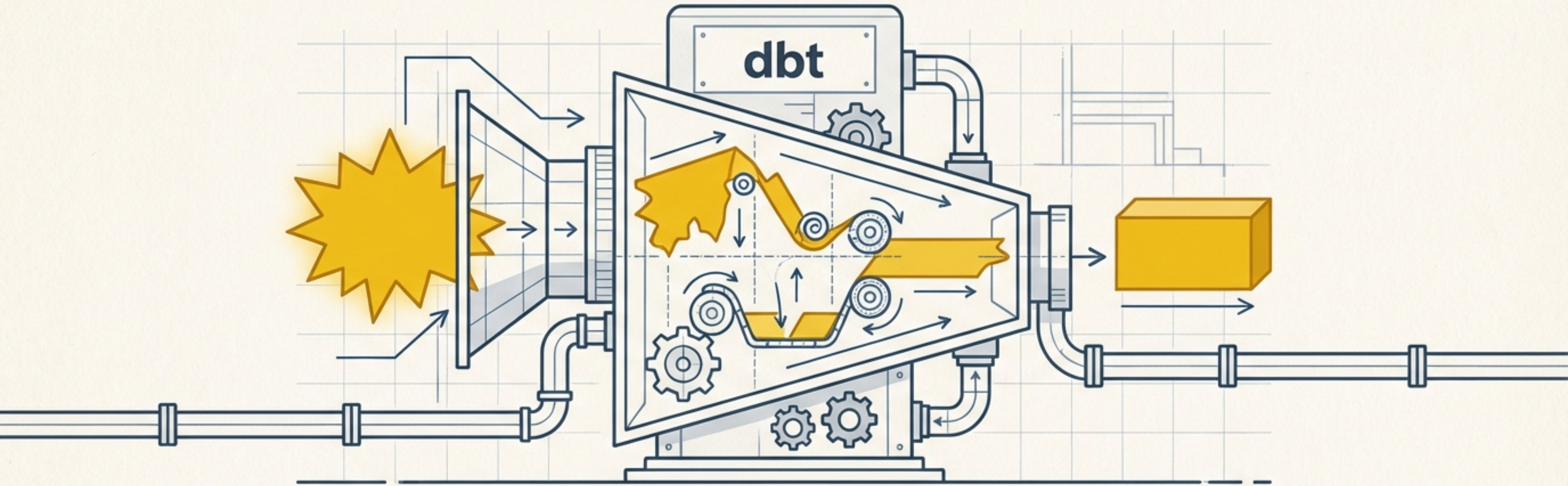
The data is not changed here; it is simply copied.
It is the digital equivalent of moving water **from a well to a reservoir**.

Arrival in the Landing Zone (Stage 1)



The data is safe in the Data Lake (Parquet files), but it is “raw.” It contains sensitive PII (Leo's Name) and is locked in a nested technical format (JSON) that is hard to analyze.

The Refinery: Making Sense of the Mess



Enter the transformation engine. Think of this as a master chef prepping ingredients—chopping, peeling, and organizing raw items into a structured mise-en-place.

Stage 2A: Normalization

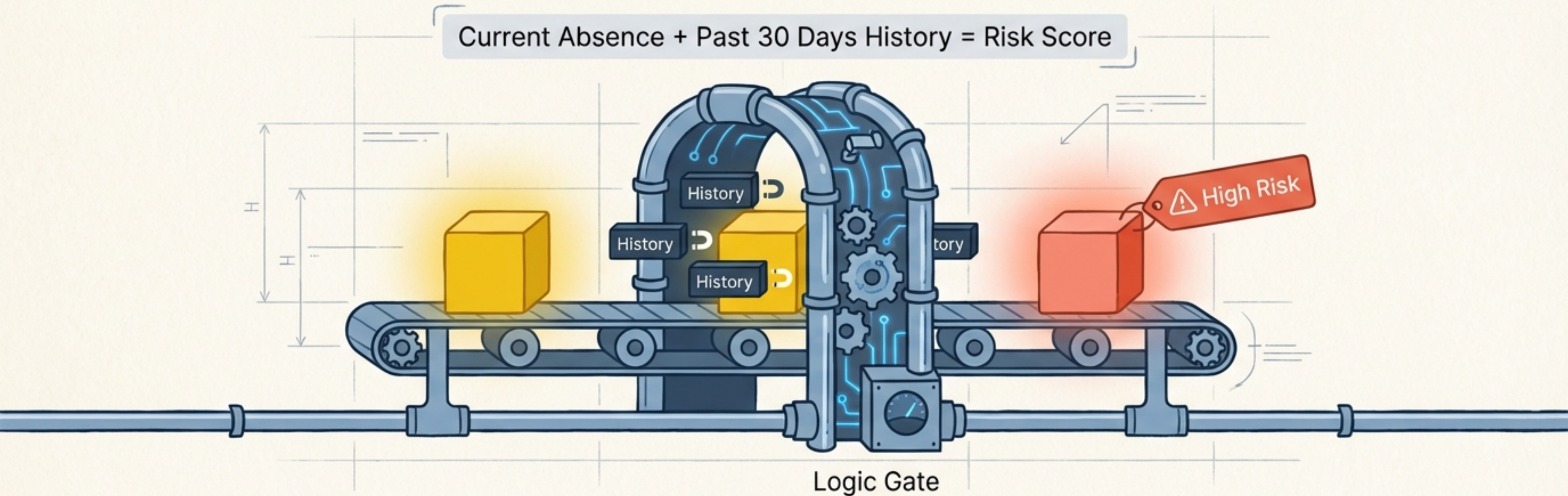
The Privacy Shield



This is **Pseudonymization**. We use a one-way hash so analysts can see that *a student* was absent, but they cannot casually see *who*.

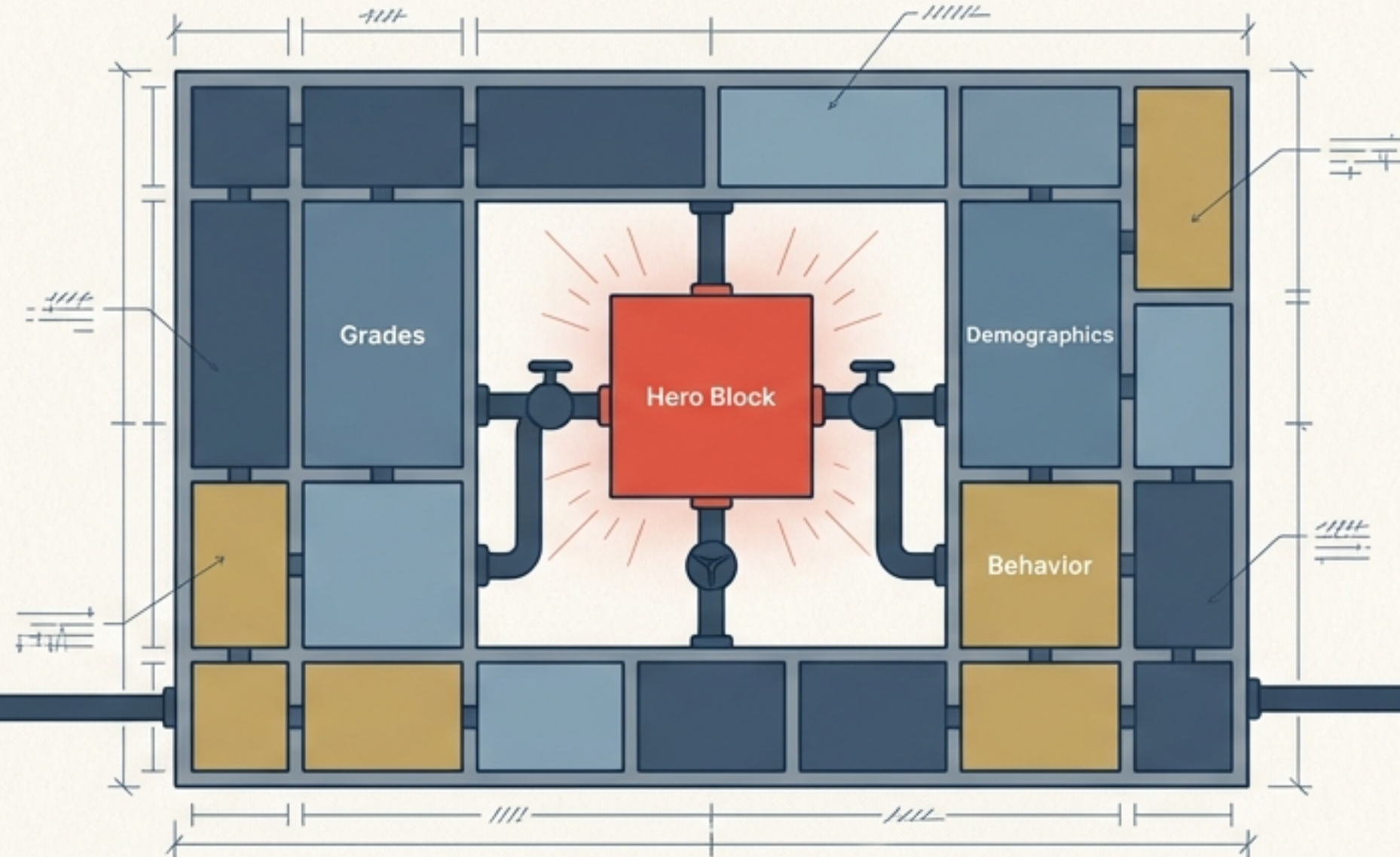
FERPA/GDPR safe. Identity is accessible only to authorized staff via a secure lookup table.

Adding Intelligence to the Record



The record stops being just "Absent" and becomes a risk indicator. By calculating history, the system flags this as "High Risk for Chronic Absenteeism".

The Gallery: Ready for the Spotlight

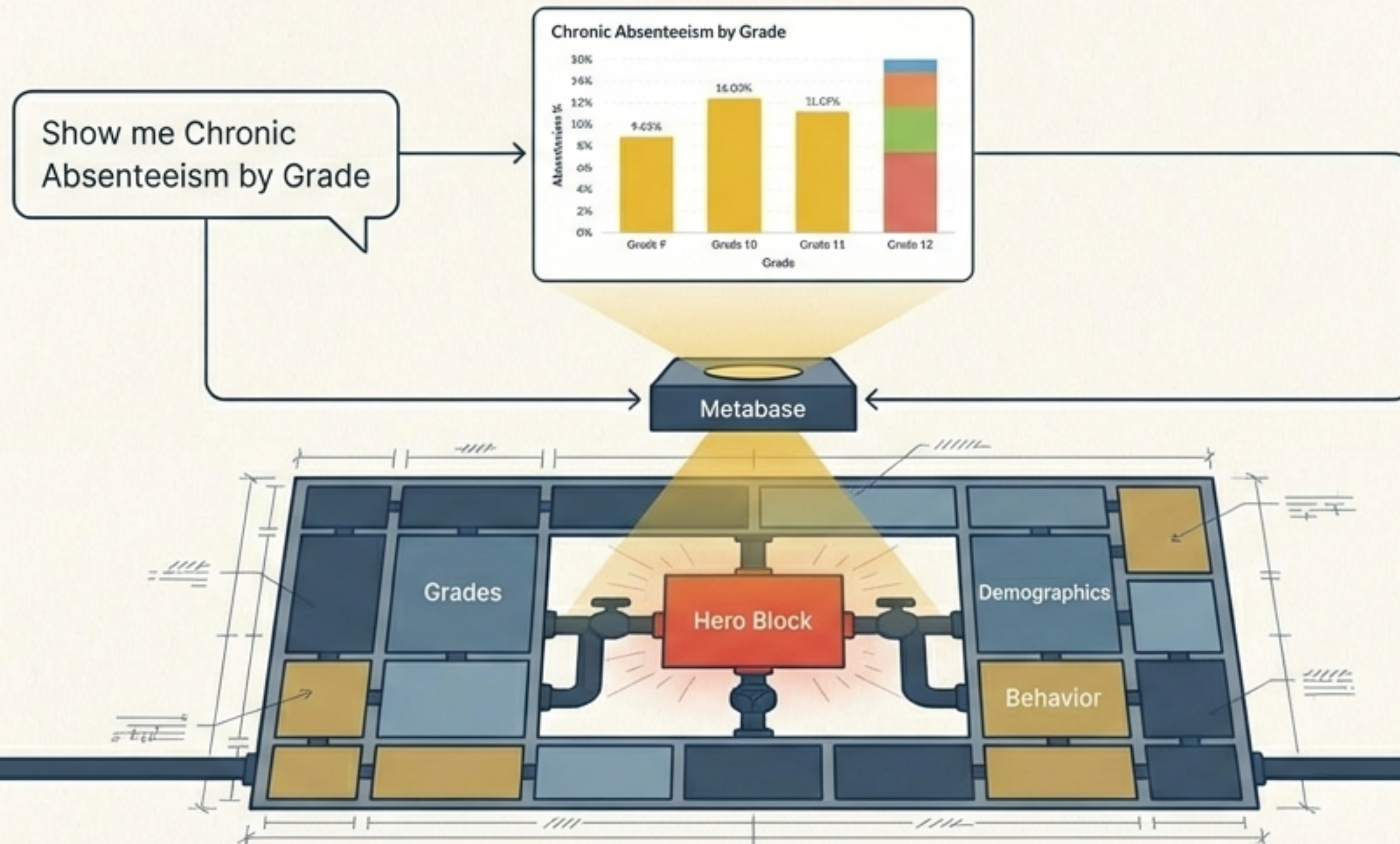


We are now in Stage 3: **Published**. The data is curated, polished, and aggregated. It is stored in **DuckDB**, a high-performance engine designed for speed (10x faster than traditional systems).

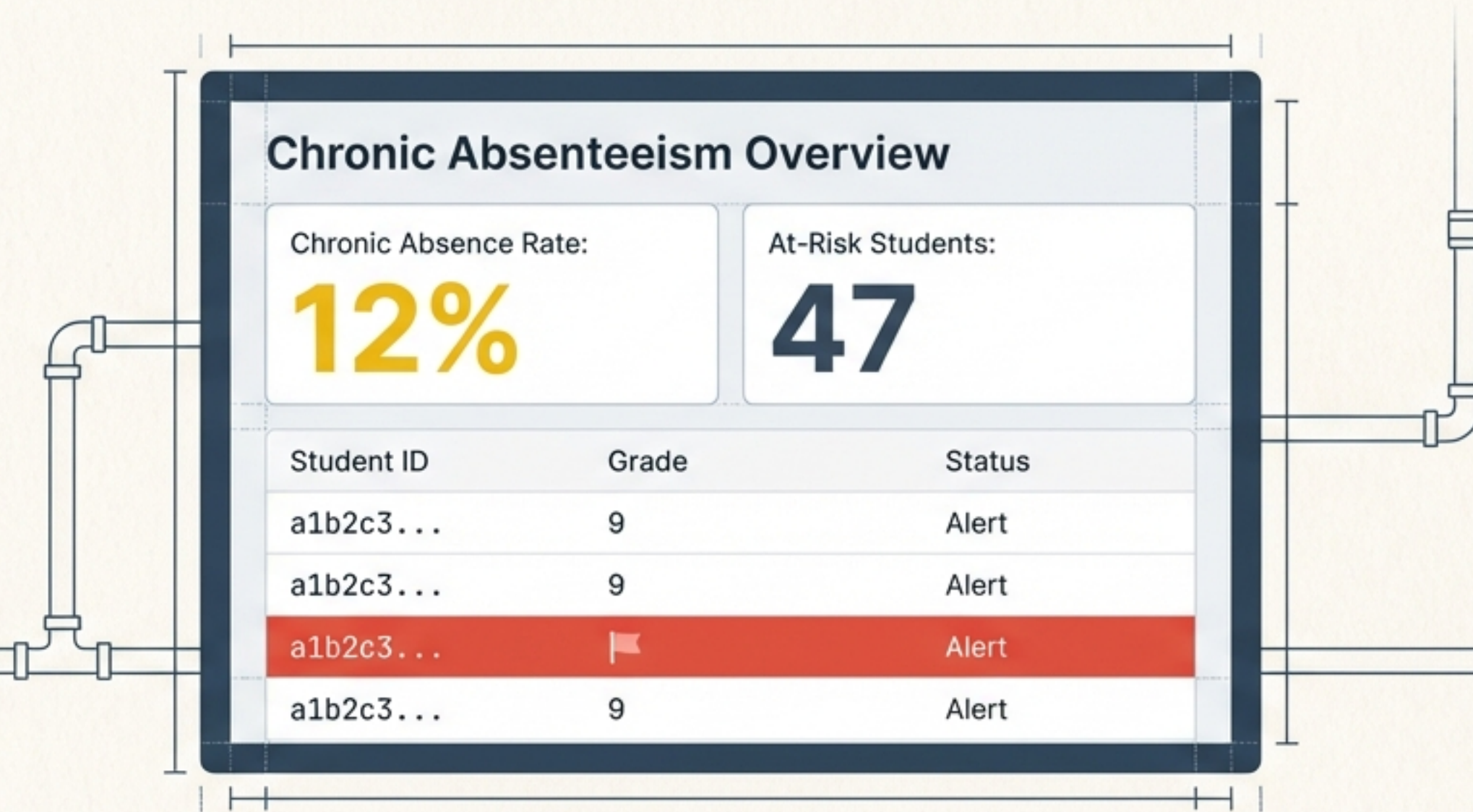
Status: Ready in the Gold Layer

Painting the Picture with Metabase

A user query triggers the scan. Because the architecture uses DuckDB and pre-computed aggregations, the chart renders in milliseconds. The text file has evolved into a visual insight.

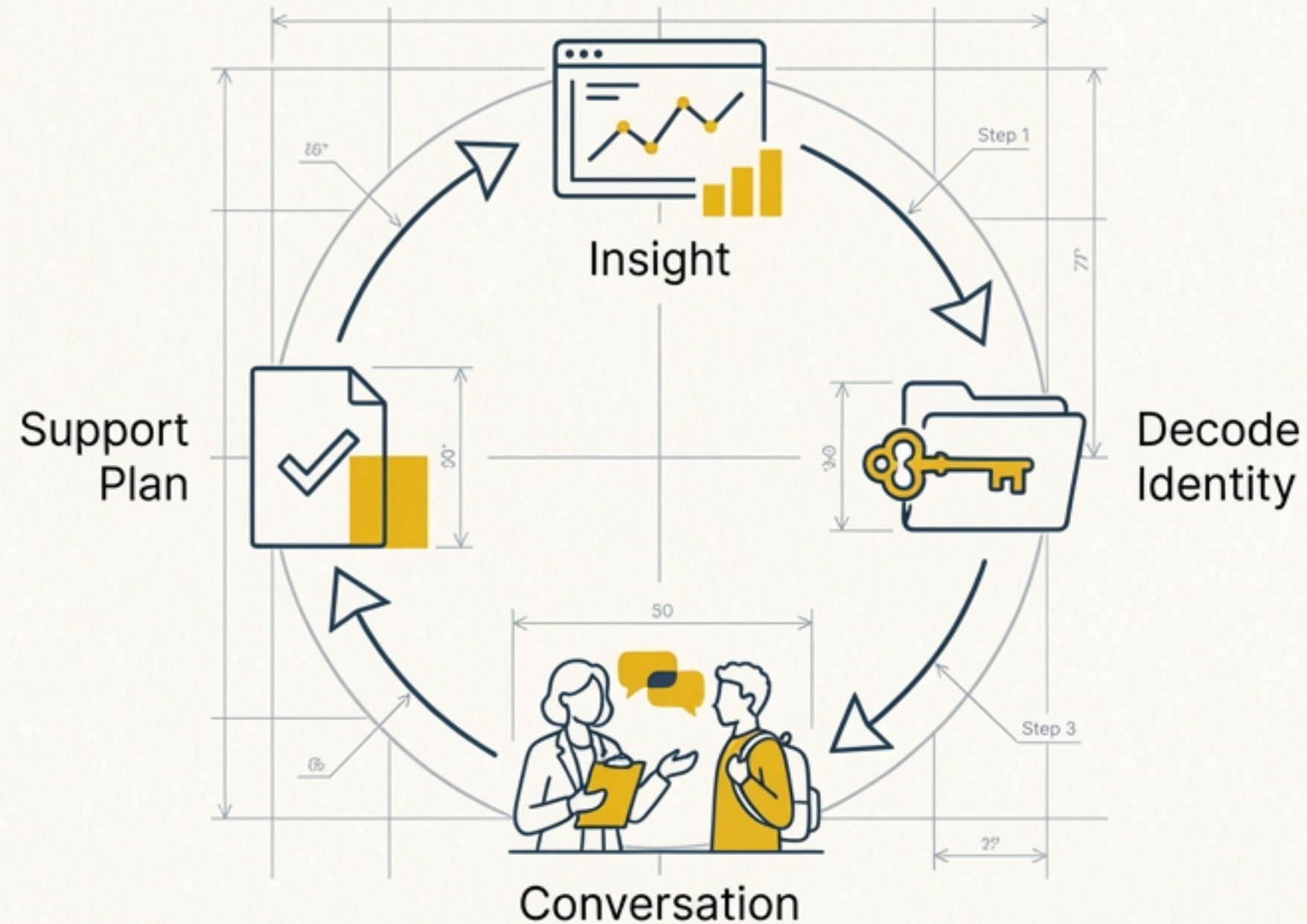


The View from the Top



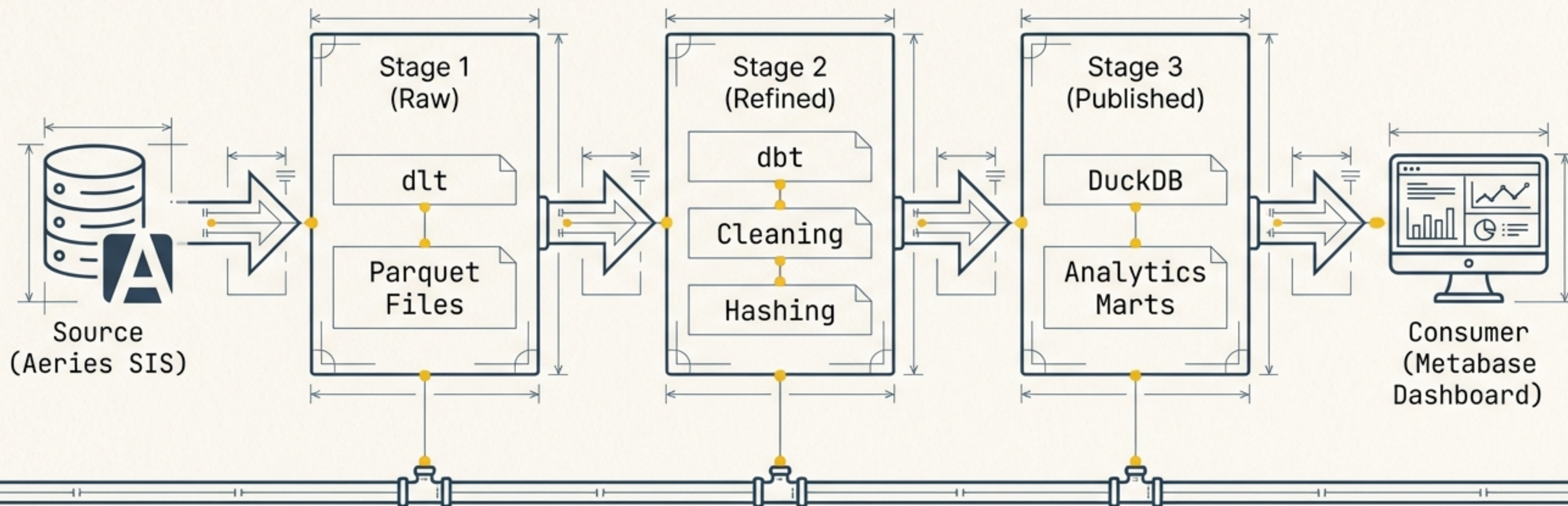
The Principal filters the view to "Grade 9" and sees the alert. The complex journey of data processing has resulted in a single, clear signal.

Closing the Loop: Data into Action



The dashboard didn't fix the problem; it revealed it.
The intervention is the true outcome.
Yesterday's absence is today's support plan.

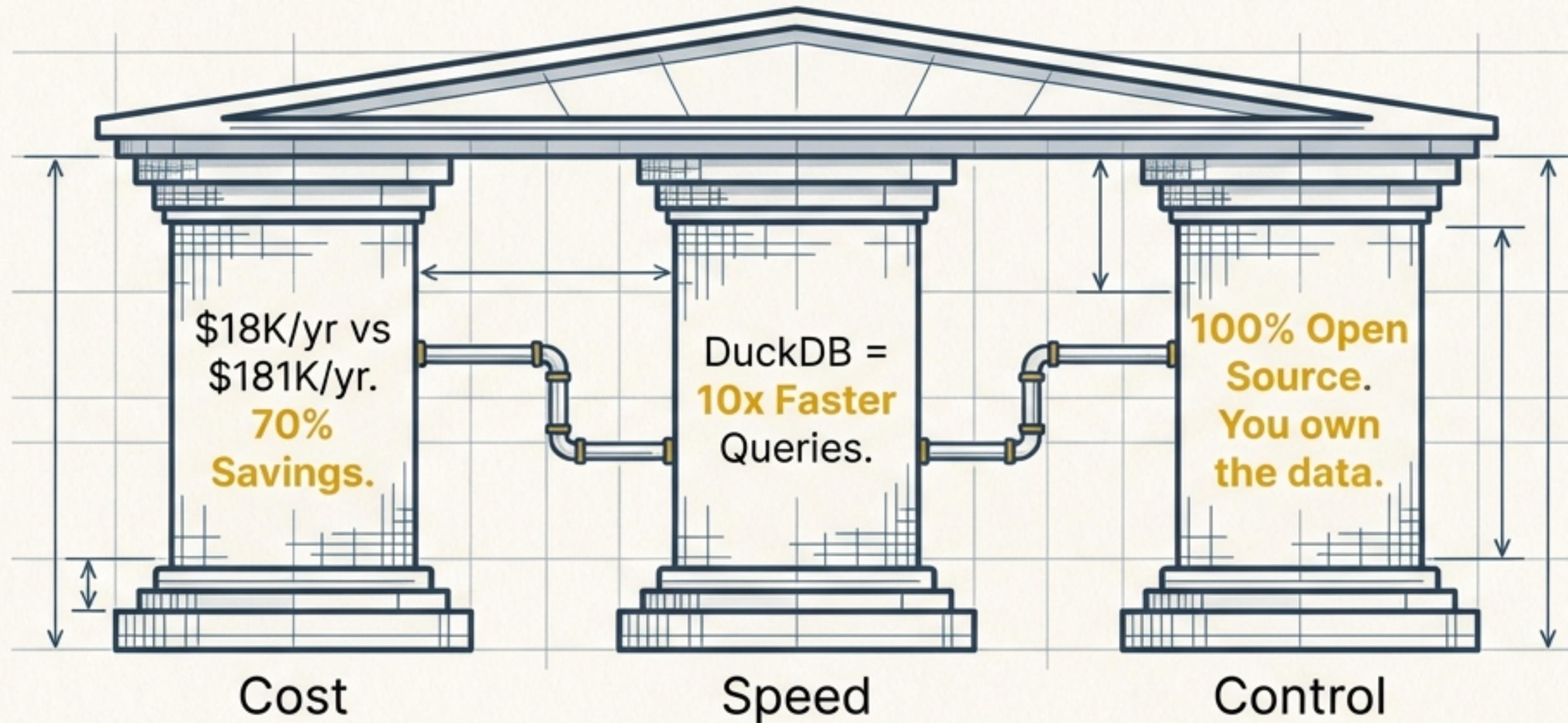
The Architecture in One View



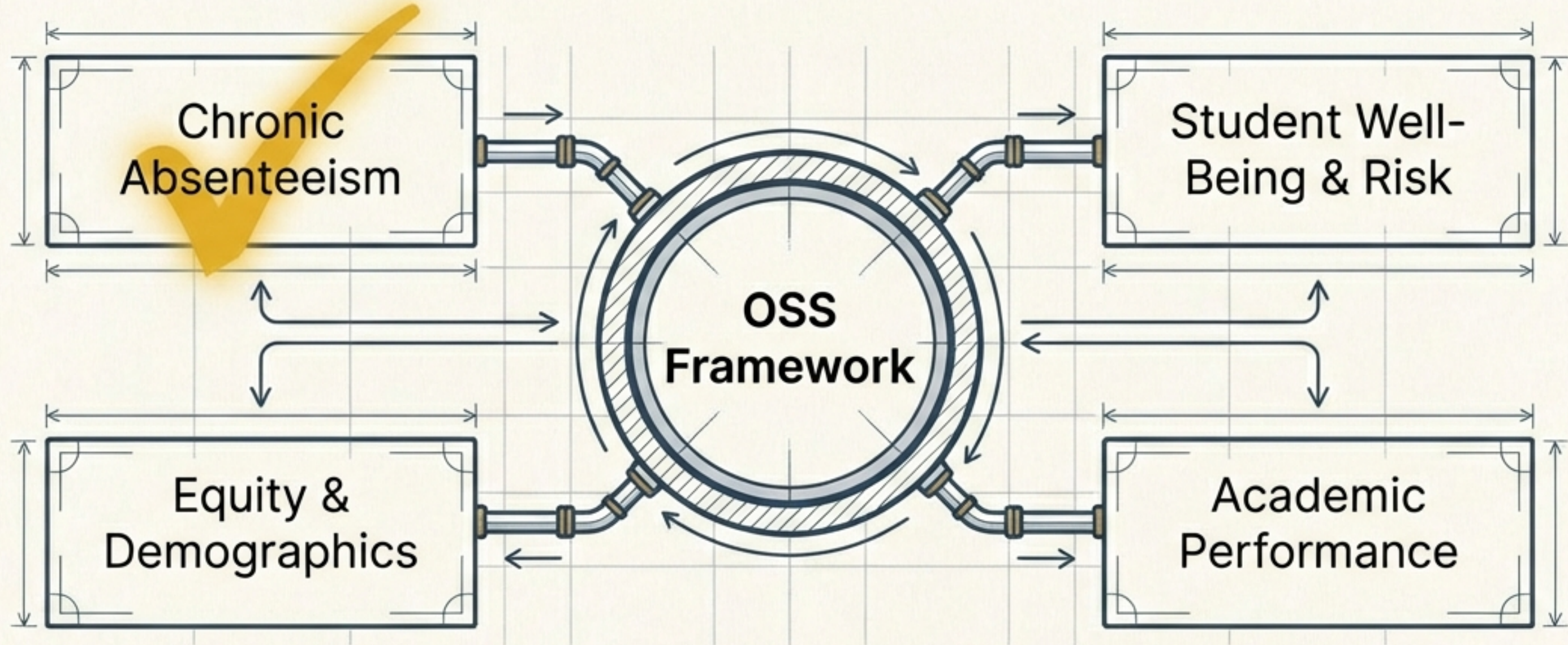
A Medallion Lakehouse architecture: moving from messy and siloed to clean and actionable.

Why This Pipeline?

The capabilities of a university data team, on a school district budget.



Beyond Attendance



The same pipeline we built for Leo works for grades, discipline, and digital equity. One framework, infinite insights.

Start Your Journey



Turn your data into your district's greatest asset.

- Explore the Chronic Absenteeism Dashboard today.
- Ask: What other questions can our data answer?